

Vidushi Toshniwal

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Education

Ph.D., Geoscience Engineering | Delft University of Technology Sep 2023-Present
Research Assistant, Geotechnical Engineering | Purdue University | GPA – 4.0/4.0 Aug 2022-Jul 2023
M.Tech, Geotechnical Engineering | Indian Institute of Technology Delhi | GPA – 8.67/10.0 Sep 2020-Jul 2022
B.Tech, Civil Engineering | Indian Institute of Technology Delhi | GPA – 8.48/10.0 Aug 2016-Jul 2020

Scholastic Achievements

- **Gerald and Beryl Leonards Fellowship:** Awarded grant worth **USD 10K** by Lyles School of Civil Engg. (2023)
- **DAAD KOSPIE:** Only student in department granted fellowship worth **INR 5.5 lakh** for research in Germany (2021)
- **Semester Exchange:** Awarded **USD 9K** by US Dept of Defense to represent IITD at Rowan University, USA (2020)
- **MHRD Scholarship, IITD:** Awarded grant worth **INR 1 lakh** by Govt. of India for exemplary research work (2020)
- **B-83 Merit Award:** Conferred amongst 2,500+ students for remarkable academic record, by IITD alumnus (2021)
- **Int'l Publication:** Toshniwal et al., 'Use of Nanomaterials in stabilisation of Bentonite Clay', IJSR IF - 7.4 (2019)
- **Summer Undergraduate Research Award (SURA):** Awarded by IRD for promising research in Geotech (2018)
- **NTSE:** Amongst top 1,000 to receive scholarship of **INR 1.26 lakh** | **NSO:** Secured AIR 106, State Rank 17 (2014)

Internships

Schlumberger Asia Services Ltd., Mumbai | *Mentor: Ankur Varshney* **Junior Field Engineer**
Drilling and measurement segment May 2019 - July 2019

- **Business Improvement:** Reduced overall tool maintenance time by 22% by designing a lean implementation strategy
- Achieved annual savings worth **USD300K** on corroded collars by developing methods for controlling rate of **corrosion**
- **Business Intelligence:** Competitive benchmarking of operational procedure, analyzed SQ issues from ONGC reports

Rowan University, USA | *Supervisor: Prof. Cheng Zhu* **Research Internship**
Noninvasive 3D imaging and characterization of permafrost profiles Jan 2020 - March 2020

- Studied relationship between electrical resistivity of frozen soil and key parameters for permafrost characterization
- Conceptualised and performed lab-scale 1D electrical resistivity measurements of frost-susceptible clayey sands, focusing on major influencing factors including initial water content, soil density, temperature, freezing-thawing

Projects

Interests: Geotechnical hazards, Offshore foundations, Soil-structure interaction, Computational geomechanics

GIS based Geotechnical Database for Collaborative GIS | INDOT Aug'22 - Apr'23

- Designed a database structure and user workflow as per the object oriented paradigm to reduce the need to perform geotechnical investigations when the data already exists
- Once implemented, this will make INDOT geotechnical office operations more efficient by saving time and resources
- Designed classes to abstract project details and developed relations between them to enable rich and complex queries

Analysing offshore foundations under cyclic loading in clay | *Prof. Tanusree* | M.Tech Project Aug'21 - Jul'22

- Project shortlisted for DAAD fellowship, analysing monopile foundations behaviour, evaluating existing p-y approaches
- Finite element analysis on PLAXIS 3D using **UDCAM-s** soil models; formulated equations for cyclic displacement degradation for millions of load cycles and validated results with the field tests available

Analysis of behaviour of deep excavations | *Prof. Ayothiraman* | B.Tech Project Jul'19 - Nov'19

- Finite Element Modeling and simulation of behaviour of underground retaining walls under dynamic loading
- Studied the existing design parameters and analysed the improvements required to make them earthquake resistant
- Achieved **33.33% decrease** in deformation by anchored wall, performed parametric analysis by varying the properties

Ground improvement by nanomaterials inclusion | *Prof. Ayothiraman* | SURA May'18 - Jan'19

- Assessed feasibility of using commercially available stabilizers to prevent damages in clay, analysed 50+ literature
- Reduced volumetric strains and crack intensity factor by conducting 13 tests on **bentonite** clay with **TerraSil**

Static and dynamic behaviour of soils and risk assessment | *Prof. Shwetha* | Summer Project May'21 - Jul'21

- Simulated seismic response using multi-hazard analysis for stochastic time domain phenomena using MOOSE app
- Developed function to analyse coupling of solids and fluids behavior, used tensor mechanics and porous flow modules

Other Projects

Numerical method of flow net sketching: Developed flow net around sheet pile wall using numerical iteration method

Embankment design: Analysed stability of slope made of pond ash and fly ash using PLAXIS after every construction and consolidation stage and performed experiments to determine the material properties of ash

Retaining wall design: Used ordinary method of slices, Bishop's and Janbu's simplified method, Janbu's chart for FOS

Flood modeling and simulation using IRIC: Studied IRIC Nays2DFlood solver to use as early flood warning system

Portable worm-bin composting machine: Designed plastic-netted metallic bin, optimized composting rate by temperature regulation, made portable bin with 3 compartments

Coursework

Relevant Courses: Engineering behaviour of soils, Site investigation and foundation, Finite element method in geotech, Ground improvement and geosynthetics, Soil dynamics and Earthquake geotechnical, Slopes and retaining structures, Geoenvironmental engineering, Mechanics of solids, Computational aspects in water resources

Purdue - Finite element method in aerospace, Continuum mechanics, Advanced foundation engineering, Plasticity theory

Laboratory: Geotechnical engineering lab, Engineering geology laboratory, Soil mechanics laboratory, Soil engineering laboratory, Geoenvironmental and geotechnical engineering laboratory

Positions of Responsibility and Extra Curriculars

Vice President, GIGSO | Geo-Institute Graduate Student Organization *Apr'23 - Jul'23*

Community and Outreach Team, CEGSAC | Civil Engineering Graduate Student Organisation *Aug'22 - Apr'23*

General Secretary, NSS | Social Service Club of IIT Delhi | Led 4-tier team of 35+ members *Apr'19 - Jun'20*

○ Initiated projects with Delhi Govt. reforming education in govt. schools; spearheaded 40+ projects, 30+ collaborations

Internal Complaints Committee | Dealing with the issues of sexual harrasement in campus *Sep'20 - Jul'22*

○ Envisioned first online cultural fest and reduced cost by 40% by introducing collaborative platform sharing strategy

Mentorship Coordinator | BSW, IIT Delhi | Implemented initiatives aiding 6000+ students *Sep'20 - Jul'21*

○ Conducted sessions on mental health impacting 2,000+ students; organised Gatekeeper Training for 200+ mentors

Extra Curriculars

Ankahi | Theatre Society of IIT Delhi | Directed and acted in 15 plays *Apr'17 - Dec'20*

○ **National Level:** Winner, General Championship Trophy at Second Inter IIT Cultural Meet hosted by IIT Kanpur

Social Endeavours | Volunteer | Contributed 4+ years, associated with 20+ NGOs *Sep'16 - Jul'20*

○ Associated with National Association for blinds and mentored 30+ visually impaired girls, organised 4 blood donation camps, organised book collection drive achieved 200% YoY increase in collection

Awards and Recognitions | IIT Delhi | Accoladed for accomplishments in extra-currics *Apr'17 - Jun'21*

○ Awarded Gem of Himadri, highest honour bestowed for exemplary achievements at institute and hostel level (2020)